

DIGITAL IMAGE PROCESSING

LECTURE # 1

INTRODUCTION

Contact Information

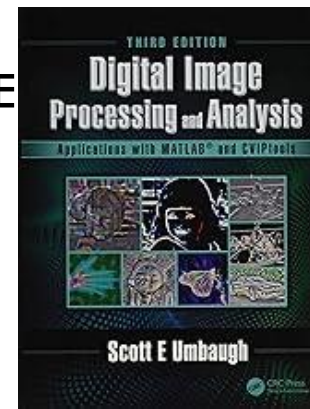
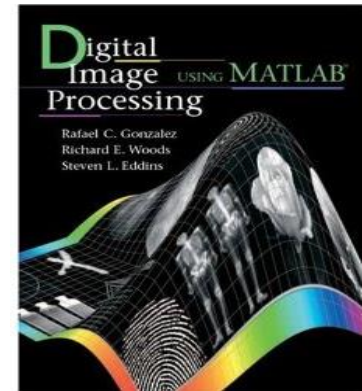
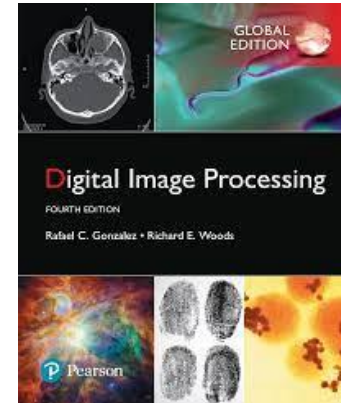
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- ❑ Course Instructor: **Dr. Ali Javed**
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- ✓ Contact No: **+92-51-9047747**
- ✓ Office hours:
 - **Monday, 9:00 - 11:00, Office # 7 S.E.D**
- ❑ Lab Instructor: **Dr. Saba**
- ❑ Course TA: **Ms. Hafsa**

Books

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- ❑ Gonzalez, R. C. and Woods, R. E., Digital Image Processing, Fourth Edition, Pearson-Prentice Hall, Inc., 2017.
- ❑ Gonzalez, R. C., Woods, R. E., and Eddins, S. L., Digital Image Processing Using MATLAB®, Pearson-Prentice Hall, Inc., 2004, ISBN 81-7758-898-2.
- ❑ Digital Image Processing and Analysis by Scott E Umbaugh, 3rd edition, 2017



Links and Reference Material

4

- ❑ <http://www.mathworks.com/index.html>
- ❑ <http://opencv.willowgarage.com/wiki/>
- ❑ <http://sourceforge.net/projects/opencvlibrary/>

Grading Criteria

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□ Grading

- ✓ Mid Exam - 25%
- ✓ Final Exam - 50%
- ✓ Quiz - 10 %
- ✓ Assignment - 15 %

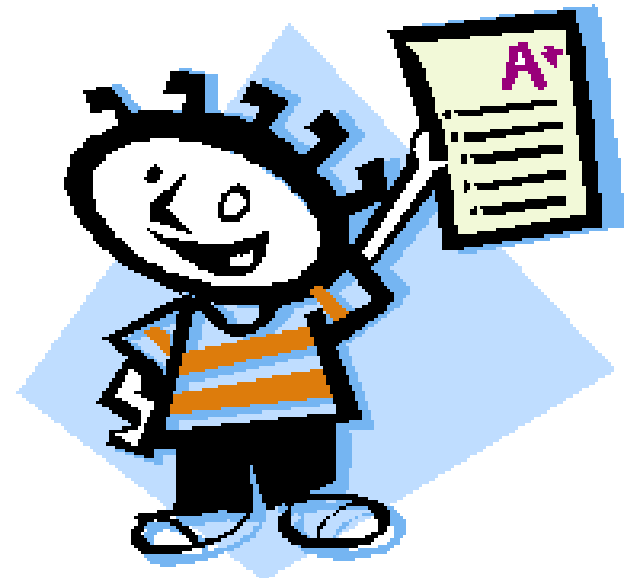
Quizzes and Assignments

6

❑ Quiz- 2-3 quizzes



❑ Assignment- 3-4 assignments



Course Outline

- ❑ Introduction to Digital Image Processing
- ❑ Applications of Digital Image Processing
- ❑ Digital Image Fundamentals
- ❑ Image enhancement in spatial domain
- ❑ Image enhancement in frequency domain
- ❑ Color Image Processing
- ❑ Image Compression
- ❑ Morphological Image Processing
- ❑ Real time Applications and Problems in DIP

Course Learning Outcomes

At the end of the course, students are expected to;

1. demonstrate in-depth knowledge of image and 2-D signal processing and use their mathematical interpretation.
2. Apply various transformation functions/filters for image enhancement, compression, morphology for binary, grayscale and color images.
3. devise and evaluate algorithms for real-time problem solving using tools like MATLAB and MS Visual Studio with OpenCV by conducting independent/ group study.

CLO Mapping

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Mapping of CLO's to taxonomy and PLO	Course Learning Outcome	*BT Level	PLO	PLO Emphasis Level
	At the end of the course, students are expected to;			
	1. demonstrate in-depth knowledge of image and 2-D signal processing and use their mathematical interpretation.	C2	1	2
	2. Apply various transformation functions/filters for image enhancement, compression, morphology for binary, grayscale, and color images.	C3	2	3
	3. devise and evaluate algorithms for real-time problem solving using tools like MATLAB and MS Visual Studio with OpenCV by conducting independent/group study.	C6, A2	3	3
	*BT Level=Bloom's Taxonomy Level C(Cognitive Domain): C1(Remembering), C2(Understanding), C3(Applying), C4(Analyzing), C5(Evaluation), C6(Creating) *PLO Emphasis Level 1=low, 2=medium, 3=high			

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Image

“A picture is worth a thousand words”

Anonymous



What is an Image?

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- ❑ Image is a source of information according to information theory
- ❑ An image may be defined as a two dimensional function $f(x,y)$ where x and y are spatial coordinates and amplitude of f at any pair of coordinates (x,y) is called the intensity or Gray level of the image at that point.



You should be able to see 11 shades of gray - if not you should adjust your monitor.



Digital Image

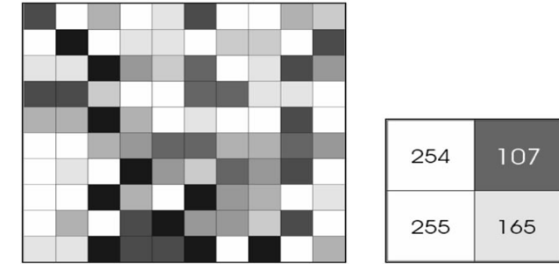
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- ❑ When x, y and the amplitude values of f are all finite, discrete quantities, we call the image a Digital Image.
- ❑ A digital Image is composed of a finite number of elements each of which has a particular location and value
- ❑ These elements are referred to as Picture Elements, Image Elements, Pels or Pixels.
- ❑ In digital imaging, a pixel is the smallest piece of information in an image.

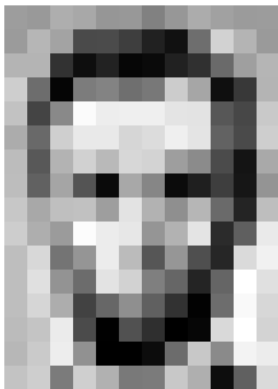


Pixel

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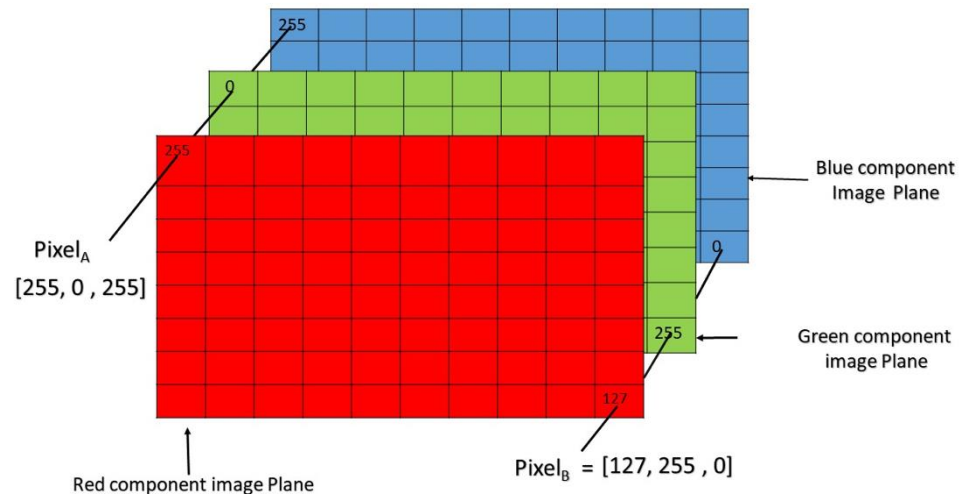


- ❑ Pixels are normally arranged in a regular 2-dimensional grid, and are often represented using dots or squares
- ❑ The intensity of each pixel is variable; in grayscale images we have one color value while in color systems, each pixel has typically three or four components such as red, green, and blue, or cyan, magenta, yellow, and black



167	153	174	168	160	162	129	161	172	161	165	166
165	182	163	74	75	62	93	17	110	210	180	154
180	180	50	14	54	6	10	93	48	195	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	194	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	118	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

167	153	174	168	160	162	129	161	172	161	165	166
165	182	163	74	75	62	93	17	110	210	180	154
180	180	50	14	54	6	10	93	48	195	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	194	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	118	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218



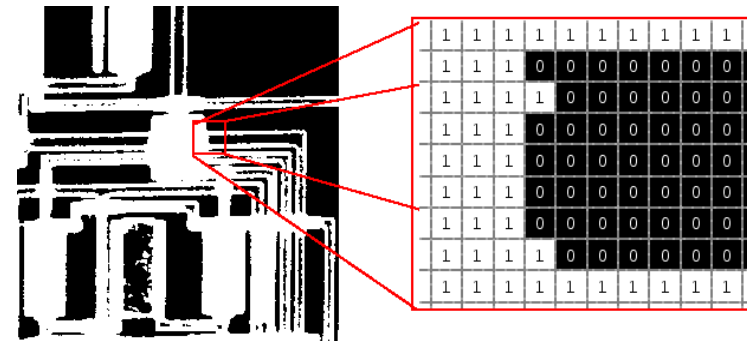
Pixel of an RGB image are formed from the corresponding pixel of the three component images

Image Types

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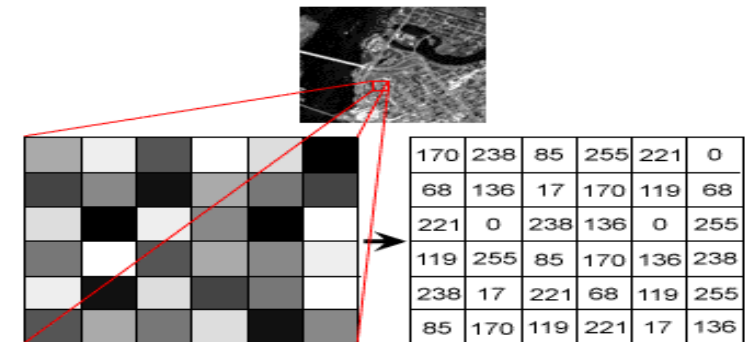
Binary Image

- ✓ 1 Sample per point



Gray scale Image

- ✓ 1 Sample per point



Color Image

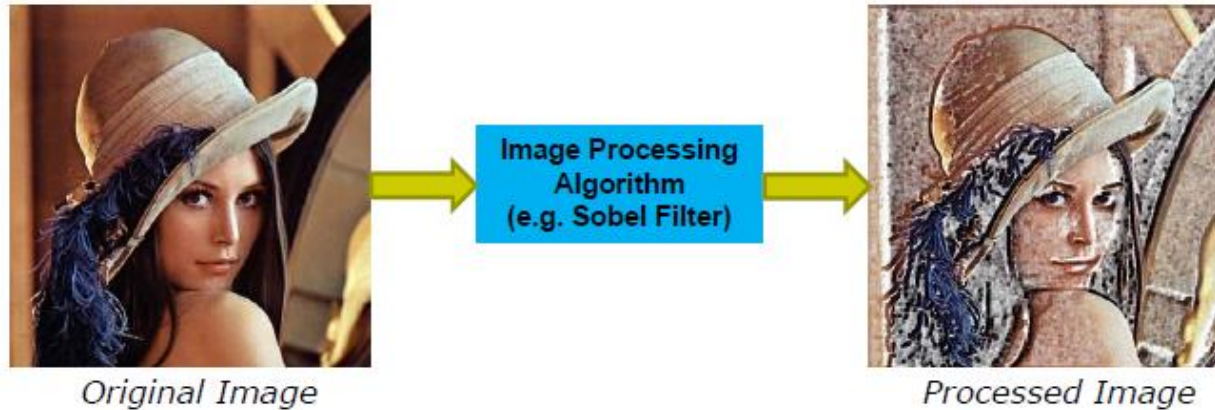
- ✓ 3 or 4 Samples per point

69	R: 68	R: 70	R: 71	R: 73	R: 76	R: 74	R: 71	R: 72	R: 76	R:
44	G: 43	G: 43	G: 44	G: 43	G: 43	G: 41	G: 42	G: 44	G: 54	G:
68	B: 70	B: 72	B: 70	B: 66	B: 65	B: 69	B: 70	B: 67	B: 61	B:
70	R: 71	R: 71	R: 69	R: 70	R: 69	R: 72	R: 87	R:110	R:128	R:1
45	G: 44	G: 44	G: 42	G: 43	G: 41	G: 46	G: 64	G: 90	G:116	G:1
73	B: 70	B: 67	B: 65	B: 67	B: 67	B: 62	B: 55	B: 51	B: 41	B:
74	R: 72	R: 70	R: 69	R: 74	R: 81	R:102	R:132	R:148	R:151	R:1
42	G: 44	G: 44	G: 43	G: 49	G: 64	G: 90	G:121	G:138	G:144	G:1
75	B: 73	B: 70	B: 68	B: 64	B: 54	B: 40	B: 30	B: 25	B: 19	B:
71	R: 72	R: 75	R: 92	R:115	R:130	R:143	R:152	R:151	R:153	R:1
44	G: 44	G: 47	G: 70	G: 96	G:118	G:133	G:140	G:143	G:148	G:1
69	B: 66	B: 62	B: 53	B: 44	B: 23	B: 11	B: 11	B: 13	B: 18	B:
68	R: 75	R:103	R:129	R:135	R:145	R:151	R:153	R:157	R:164	R:1
44	G: 56	G: 89	G:120	G:126	G:135	G:141	G:143	G:145	G:150	G:1
62	B: 53	B: 47	B: 42	B: 27	B: 15	B: 7	B: 8	B: 15	B: 15	B:
78	R:115	R:136	R:131	R:142	R:151	R:153	R:157	R:160	R:164	R:1
59	G:102	G:128	G:126	G:133	G:142	G:143	G:146	G:150	G:155	G:1
48	B: 45	B: 37	B: 12	B: 11	B: 18	B: 13	B: 10	B: 18	B: 30	B:
118	R:135	R:128	R:141	R:153	R:151	R:153	R:160	R:163	R:165	R:1
105	G:127	G:124	G:133	G:143	G:143	G:145	G:151	G:154	G:154	G:1
40	B: 29	B: 14	B: 7	B: 7	B: 6	B: 9	B: 19	B: 24	B: 21	B:

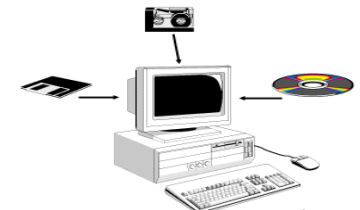
Digital Image Processing

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- ❑ Image Processing deals with algorithms that transform an input image into a new image (processed image)
- ❑ DIP is the field of processing digital images by means of a digital computer



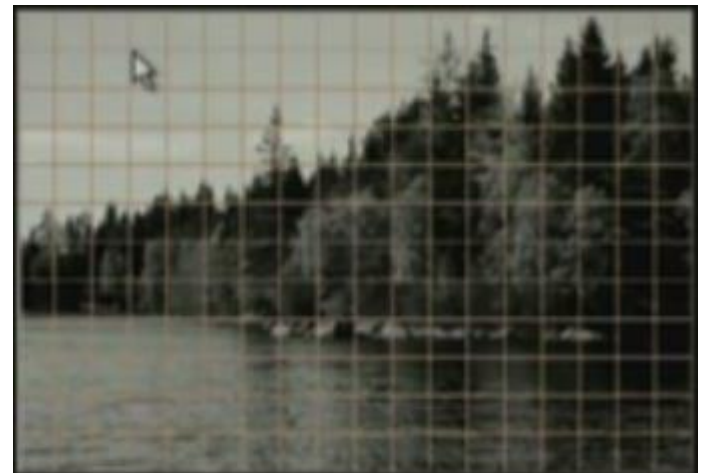
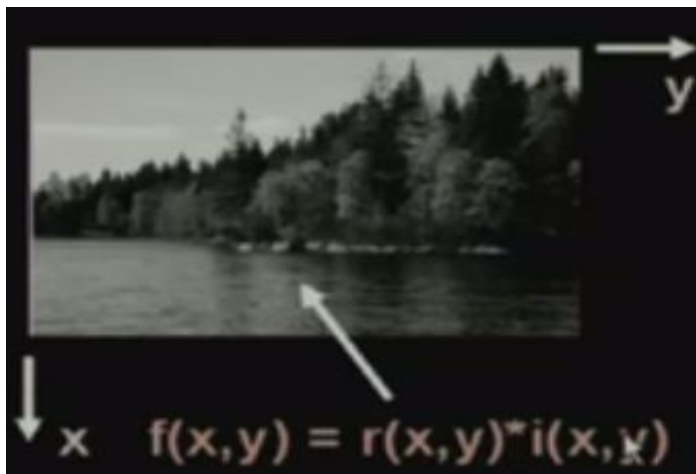
- ❑ Digital image processing focuses on two major tasks
 - ✓ Improvement of pictorial information for human interpretation
 - ✓ Processing of image data for storage, transmission and representation for autonomous machine perception



Digital Image Representation

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- ❑ A digital image $f(x,y)$ is discretized both in spatial coordinates and brightness
- ❑ It can be considered as a matrix whose row column indices specify a point in the image and the element value identifies the gray level value at that point
- ❑ Spatial discretization by Sampling
- ❑ Intensity discretization by quantization



Digital Image Representation

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Digital image is expressed as

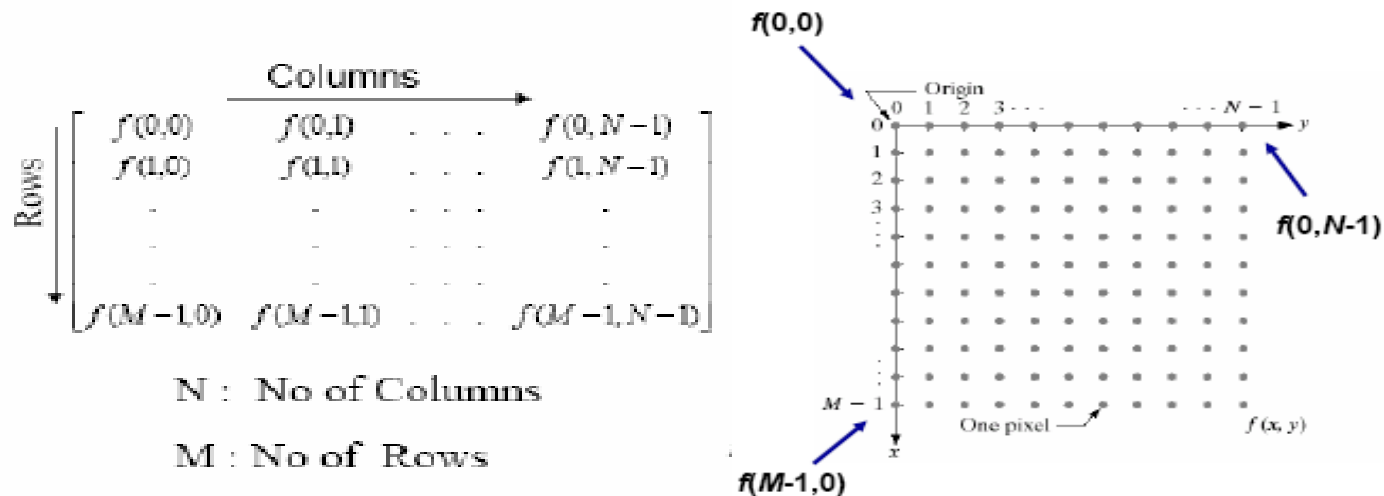
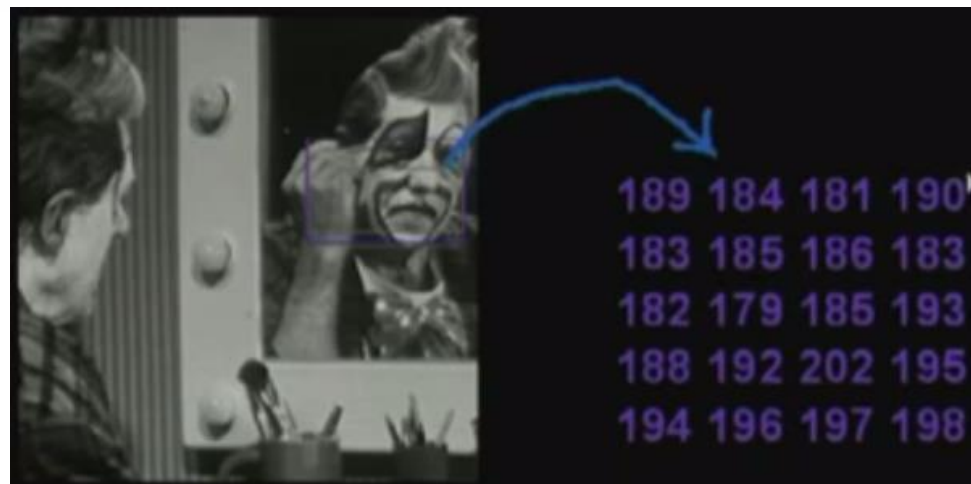


FIGURE 2.18
Coordinate convention used in this book to represent digital images.

L : # of discrete gray levels

$L = 2^k$

$b = M \times N \times k$



Video [8]

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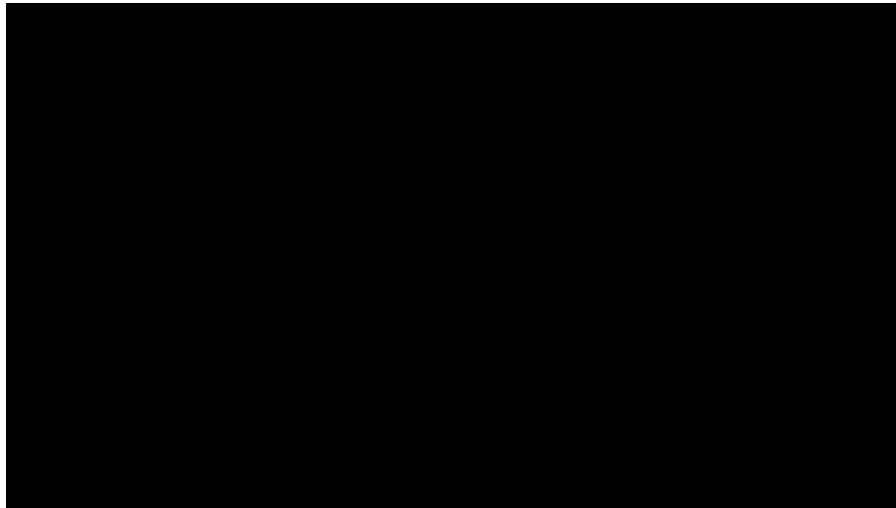
- ❑ A digital video consists of **frames** that are presented to the viewer's eye in rapid succession to create the impression of movement.
- ❑ Each frame within a digital video can be uniquely identified by its **frame index**, a serial number.
- ❑ A **shot** is a sequence of frames shot uninterruptedly by one camera. In the context of shot transition detection they are usually group into two types:
 - ✓ **Abrupt Transitions**
 - ✓ **Gradual Transitions**



Video Transitions [8]

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- ❑ **Abrupt Transitions (Cuts)** - This is a sudden transition from one shot to another, i. e. one frame belongs to the first shot, the next frame belongs to the second shot. They are also known as hard cuts or simply cuts.

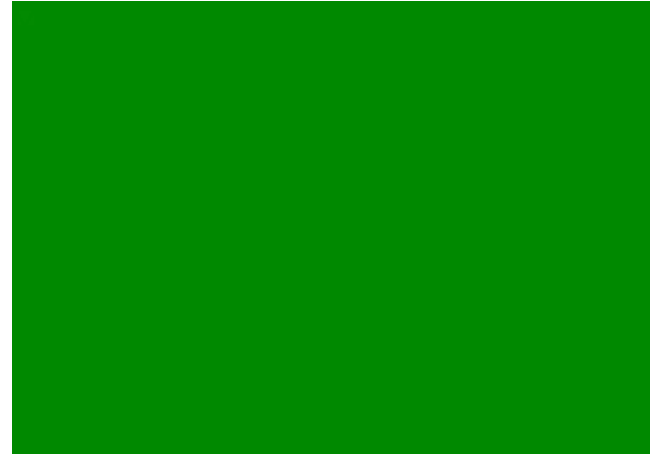


- ❑ **Gradual Transitions** - In this kind of transitions the two shots are combined using chromatic, spatial or spatial-chromatic effects which gradually replace one shot by another. These are also often known as soft transitions and can be of various types, e.g., wipes, dissolves, fades...

Gradual Transitions [9-10]

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- ❑ Dissolve: A **dissolve** is a gradual transition from one image to another.
- ❑ The terms **fade-out** (also called **fade to black**) and **fade-in** are used to describe a transition to and from a blank image.
- ❑ Wipe: A **wipe** is a type of film transition where one shot replaces another by travelling from one side of the frame to another or with a special shape.



Why Image/Video?

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- ❑ The magic of Tele-Vision (Greek word, Tele means far away, vision is for sight)
 - ✓ Our vision capability is extended in space. You don't need to travel to liberty island NY to watch statue of liberty



A Historical Overview of DIP

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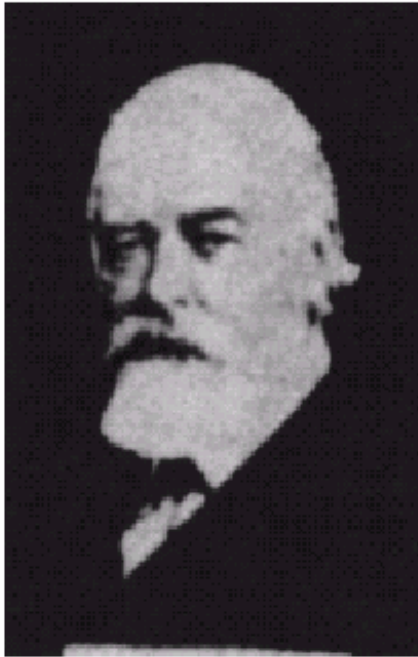
❑ **Bartlane cable picture transmission system** was a technique invented in 1920 to transmit images over cable lines between London and New York in 1920s. It was named after its inventors Harry G. Bartholomew and Maynard D. McFarlane and was first used to transmit a picture across the Atlantic in 1920. Using the Bartlane system, images could be transmitted across the Atlantic in less than three hours. [2]



A Historical Overview of DIP

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- The number of distinct gray levels coded by the Bartlane system was improved from 5 to 15 by the end of 1920s [2].



The Boom of Digital Images in the Last 25 Years

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❑ Acquisition

- ✓ Digital cameras, scanners
- ✓ Infrared and microwave imaging etc

❑ Transmission

- ✓ Internet, satellite and wireless communication

❑ Storage

- ✓ CD/DVD, Blu-ray
- ✓ Flash memory

❑ Display

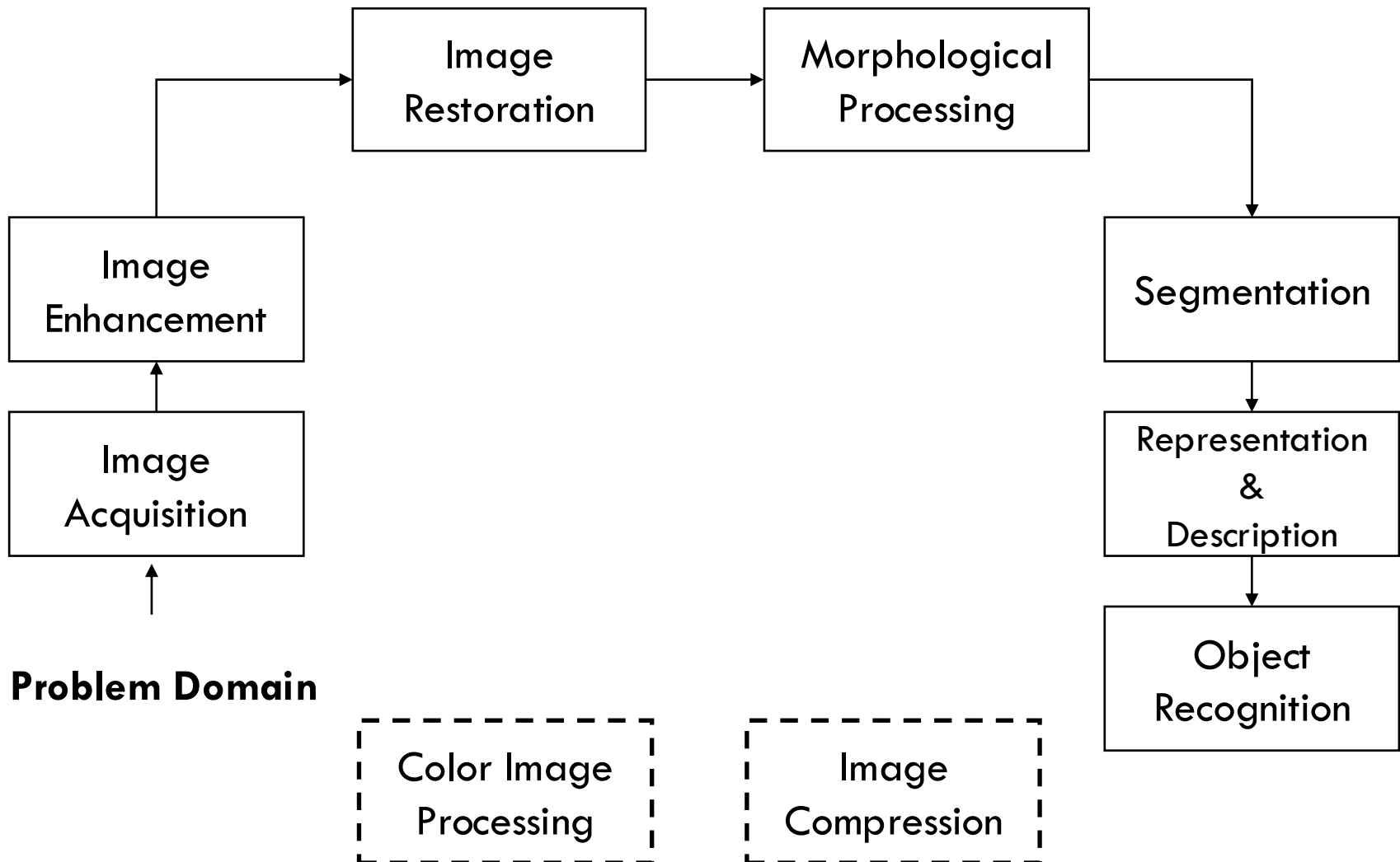
- ✓ CRT monitors, LCD monitor, LED Monitors
- ✓ PDAs, smart phones, smart watches

Key Stages in DIP

- ❑ Image Acquisition
- ❑ Image Enhancement
- ❑ Image Restoration
- ❑ Image Compression
- ❑ Color Image Processing
- ❑ Morphological Image Processing
- ❑ Image Segmentation
- ❑ Representation and Description
- ❑ Image Recognition

Key Stages in Digital Image Processing

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Key Stages in Digital Image Processing

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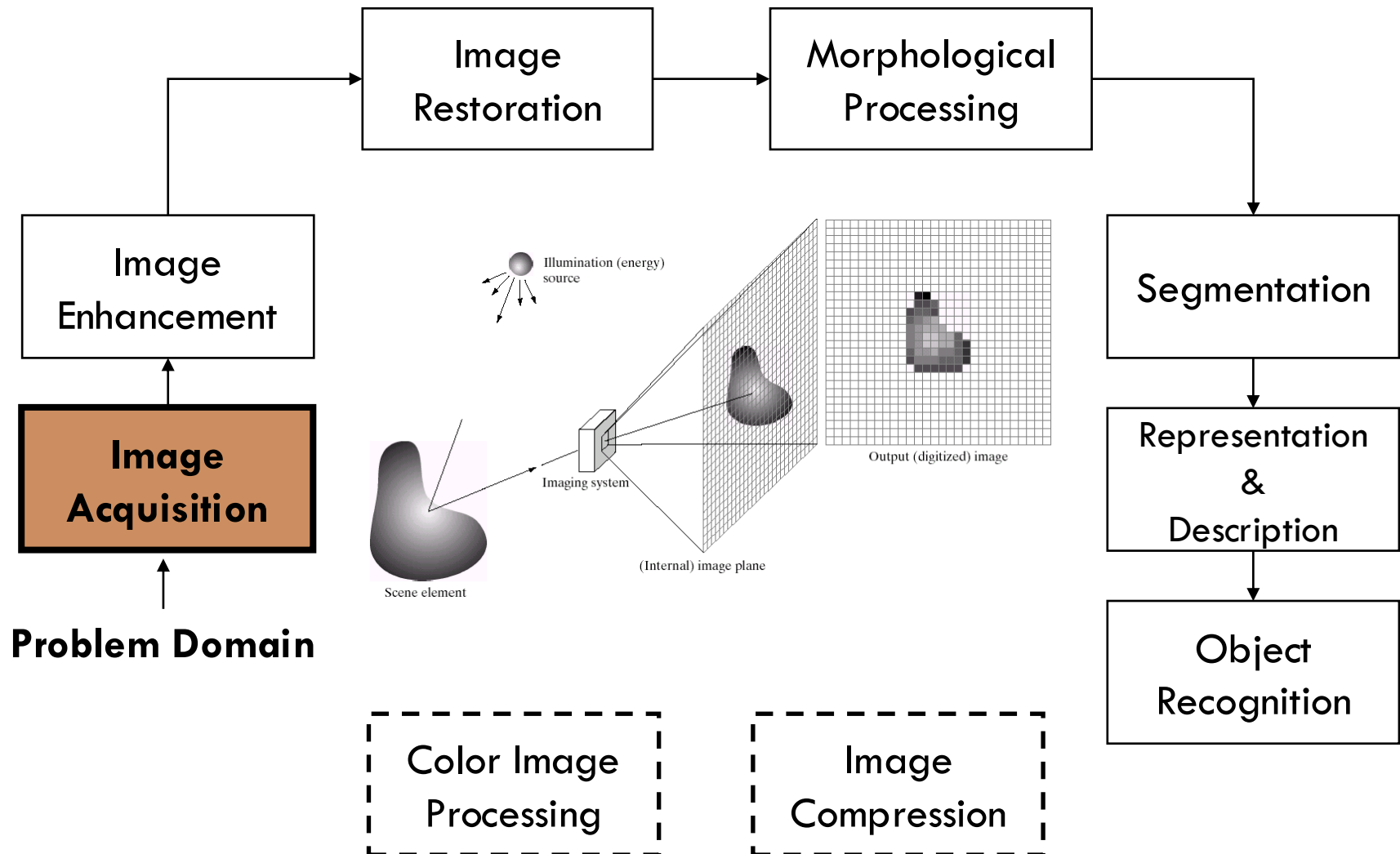


Image Acquisition

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- ❑ The first stage of any vision system is the image acquisition stage.
- ❑ An image is captured by a sensor (such as a monochrome or color TV camera) & digitized
- ❑ If the output of the camera or sensor is not already in digital form, an ADC converter digitizes it
- ❑ Images are processed after acquisition.
- ❑ However, if the image has not been acquired satisfactorily then the intended tasks may not be achievable



Key Stages in Digital Image Processing

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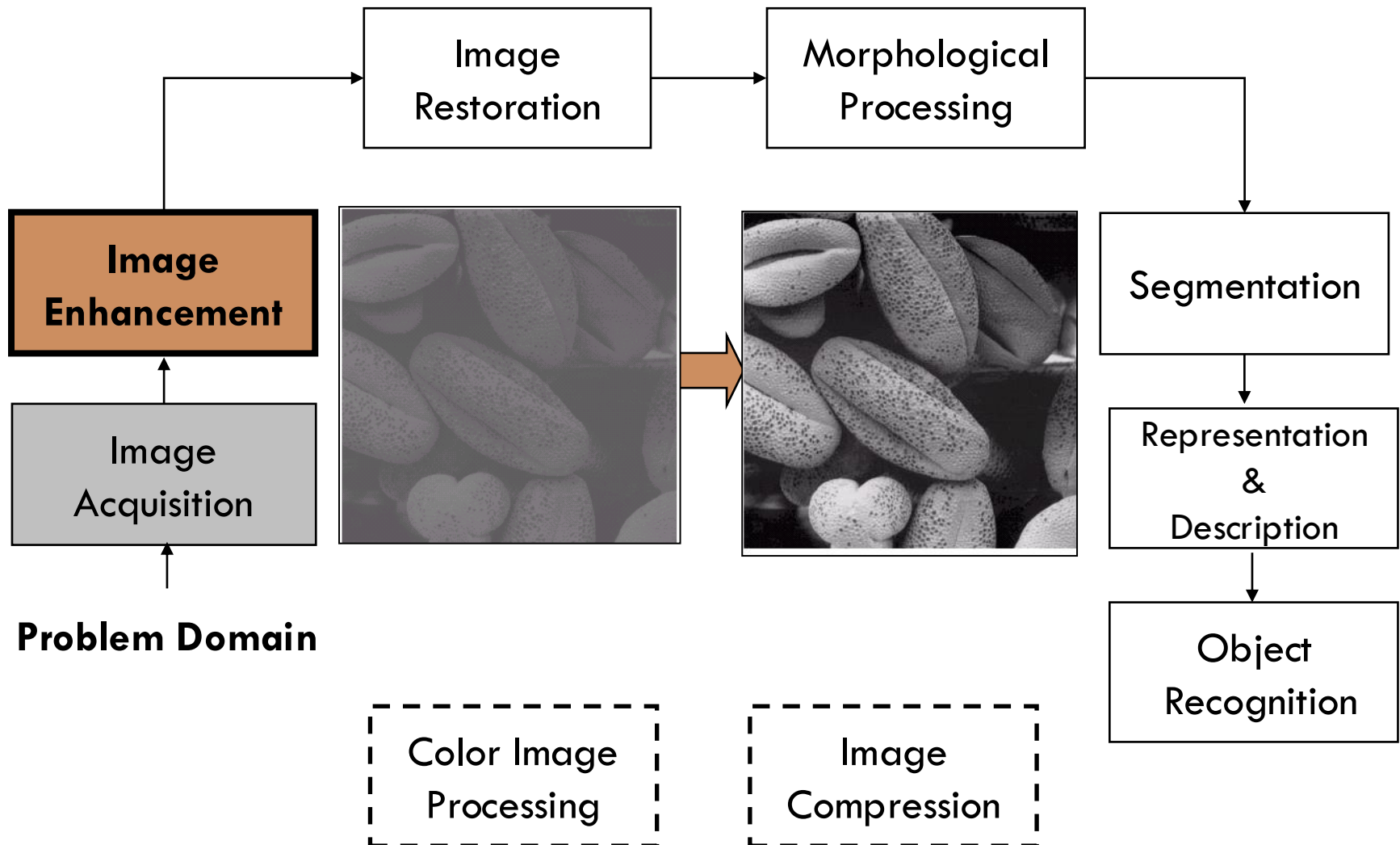


Image Enhancement

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- The aim of image enhancement is to improve the perception of information in images for human viewers, or to provide better input for other automated image processing techniques.
- Image enhancement techniques can be divided into two broad categories:
 - ✓ Spatial domain methods, which operate directly on pixels, and
 - ✓ Frequency domain methods, which operate on the Fourier transform of an image.

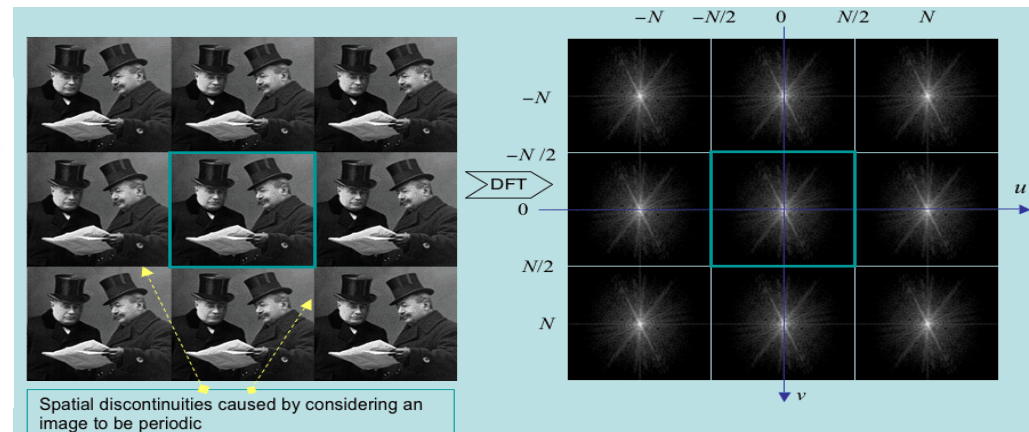
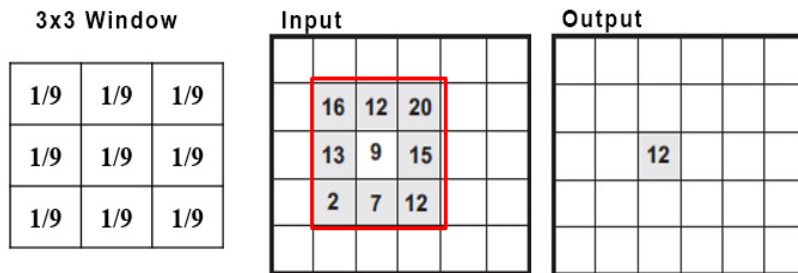


Image Enhancement

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Original Images

Enhanced Images

Key Stages in Digital Image Processing

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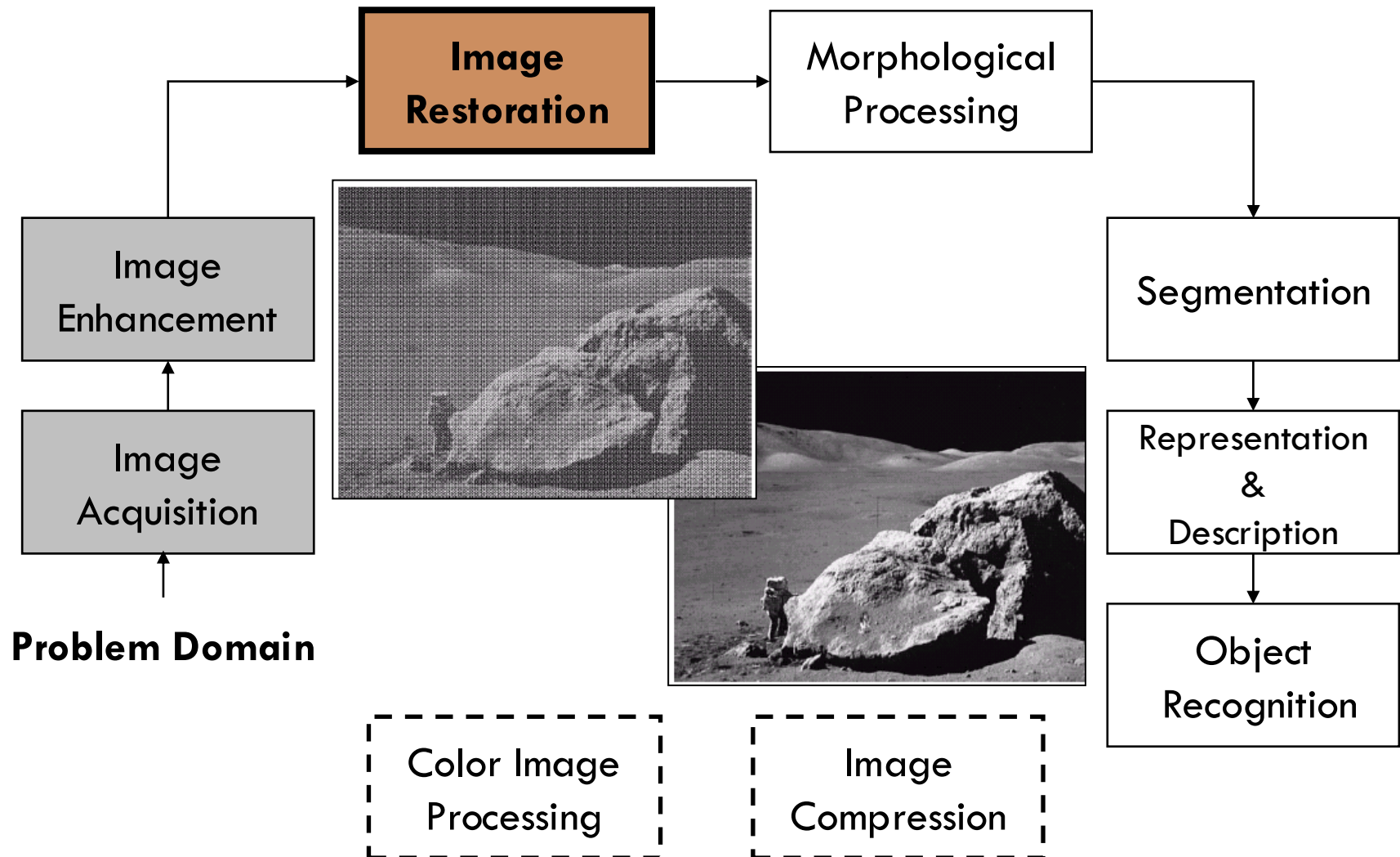


Image Restoration

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- Image restoration refers to the recovery of an original signal from degraded observations.
- The purpose of image restoration is to "compensate for" or "undo" defects which degrade an image. Degradation comes in many forms such as motion blur, noise, and camera misfocus.



Damaged Image



Restored Image

Image Enhancement vs Image Restoration

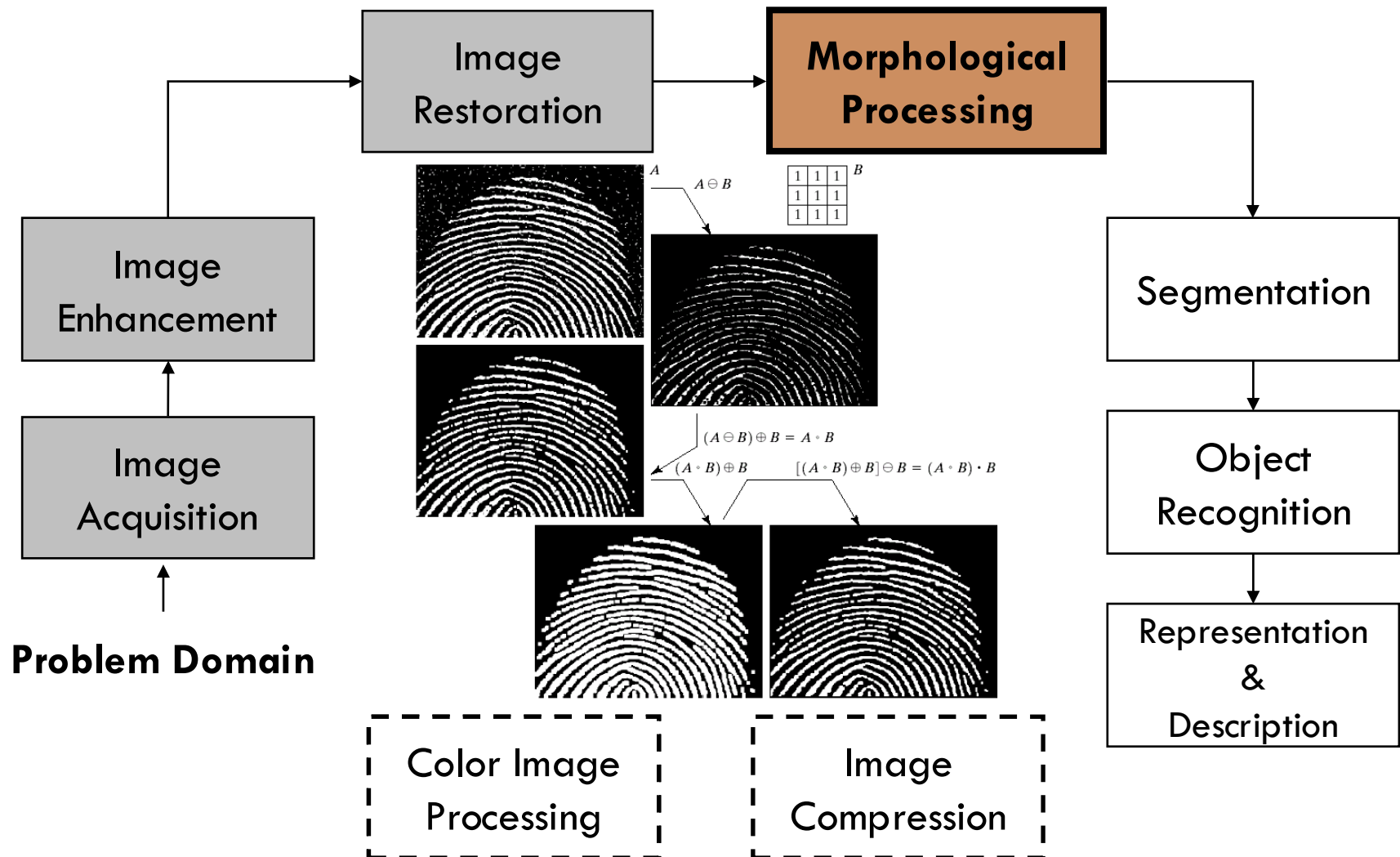
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- ❑ Image enhancement : process image to emphasize features of the image that make the image more pleasing to the observer or to process image so that the result is more suitable for a specific application, is largely a **subjective process**.
- ❑ Image restoration : recover image from distortions to its original image, is largely an **objective process**.
- ❑ Image enhancement is the improvement of digital image quality without knowledge about the source of degradation. If the source of degradation is known, one calls the process image restoration



Key Stages in Digital Image Processing

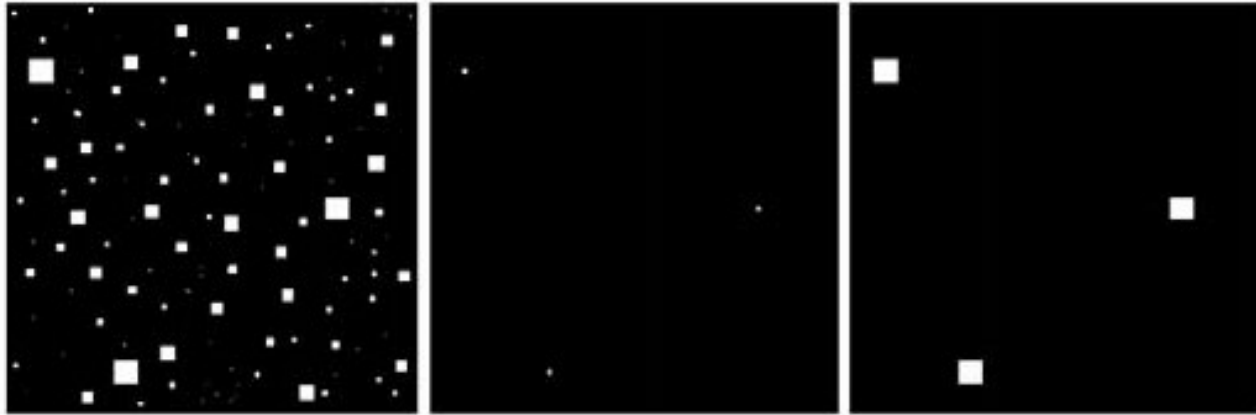
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Morphological Image Processing

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- ❑ Deals with Tools for extracting image components that are useful in the representation & description of shape



Key Stages in Digital Image Processing

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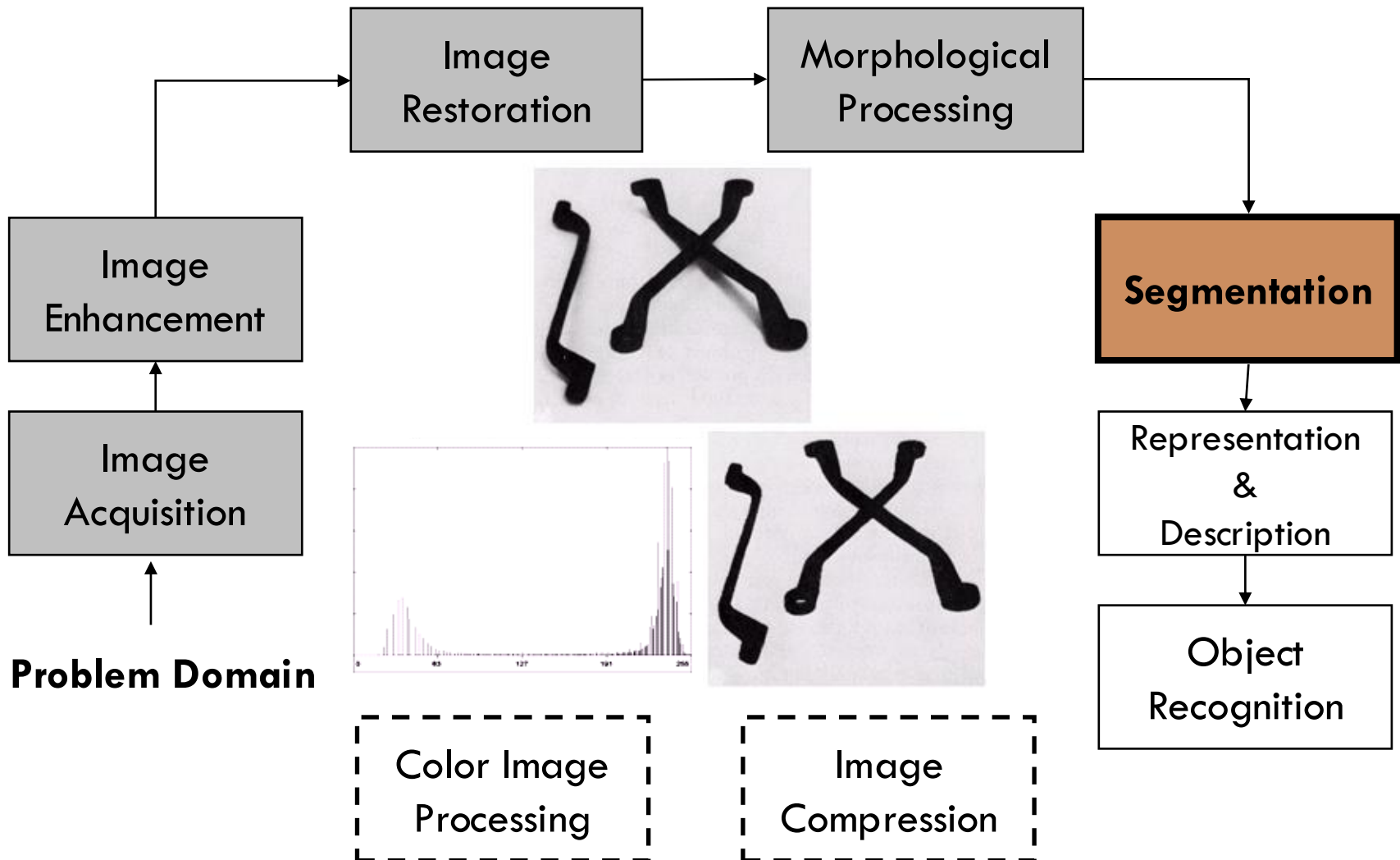


Image Segmentation



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- ❑ Segmentation refers to the process of partitioning a digital image into multiple segments (sets of pixels, also known as super pixels). The goal of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze
- ❑ Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images

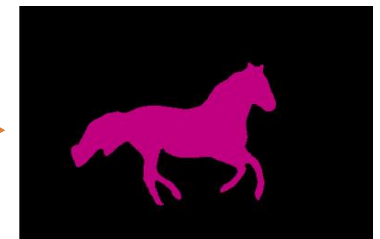
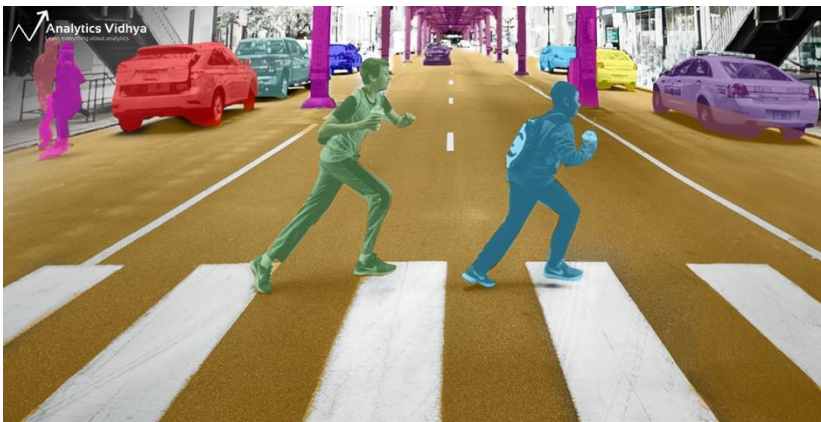
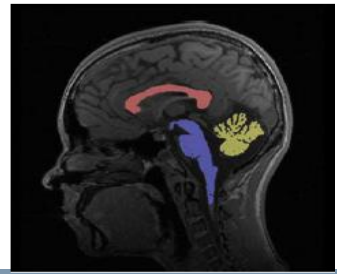


Image Segmentation



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- ❑ Object detection builds a bounding box corresponding to each class in the image. But it tells us nothing about the shape of the object. We only get the set of bounding box coordinates. We want to get more information – this is too vague for our purposes.
- ❑ Image segmentation creates a pixel-wise mask for each object in the image. This technique gives us a far more granular understanding of the object(s) in the image.

**Object
Detection**



**Instance
Segmentation**



Image Segmentation

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- ❑ Image Segmentation algorithms generally are based on one of two basic properties of intensity values:: **Discontinuity and Similarity**
- ❑ Through Discontinuity the approach is to partition an image based on abrupt changes in intensity, such as edges in an image
- ❑ Through Similarity, the approach is based on partitioning an image into regions that are similar according to a set of predefined criteria. Thresholding, region growing, region splitting and merging are examples of methods in this category

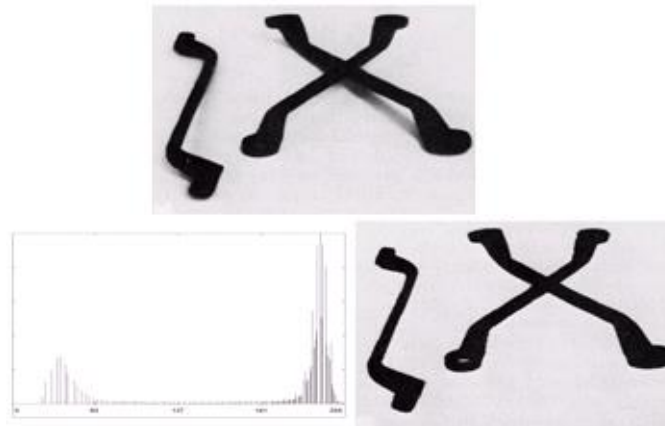


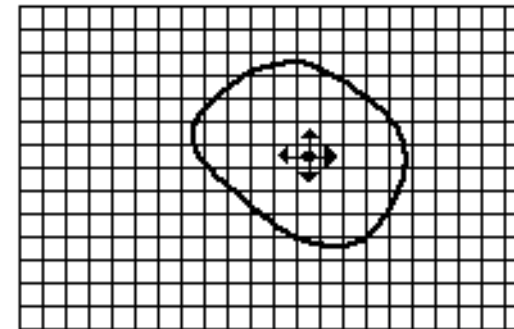
Image Segmentation

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Region growing

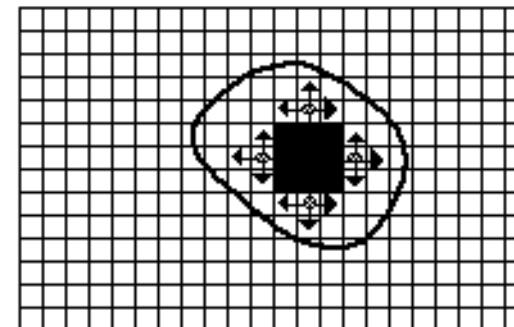
- Selection of seed point, lets select 6
- Select the threshold, lets select $t < 3$

5	6	6	7	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4



(a) Start of Growing a Region

- Seed Pixel
- ↑ Direction of Growth



(b) Growing Process After a Few Iterations

- Grown Pixels
- Pixels Being Considered

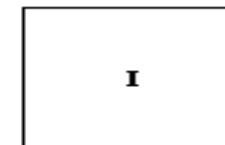
Image Segmentation

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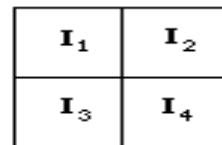
Region splitting and merging

- ❑ Split the image, e.g. in four quadrants
- ❑ Select any region, and take a difference between the maximum and minimum value in the region
- ❑ Compare the difference against the selected threshold, e.g. $t \leq 3$
- ❑ If the difference is within the threshold, then don't split the region further else split the region again into four quadrants
- ❑ Once further split is not possible then we start merging. Here we consider adjacent regions.
- ❑ Take the maximum of one region and minimum of second region and compare the difference against the threshold. Repeat this by picking the minimum of first and maximum of second region. If both meets the threshold criteria, then we can merge the two selected regions.

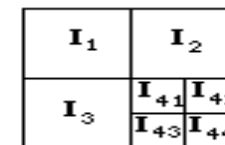
5	6	6	7	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4



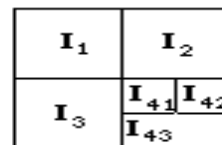
(a) Whole Image



(b) First Split



(c) Second Split



(d) Merge

Image Segmentation

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- ❑ Can you identify the difference between these two? Both the images are using image segmentation to identify and locate the people present.
- ❑ In image 1, every pixel belongs to a particular class (either background or person). Also, all the pixels belonging to a particular class are represented by the same color (background as black and person as pink). This is an example of semantic segmentation
- ❑ Image 2 has also assigned a particular class to each pixel of the image. However, different objects of the same class have different colors (Person 1 as red, Person 2 as green, background as black, etc.). This is an example of instance segmentation.



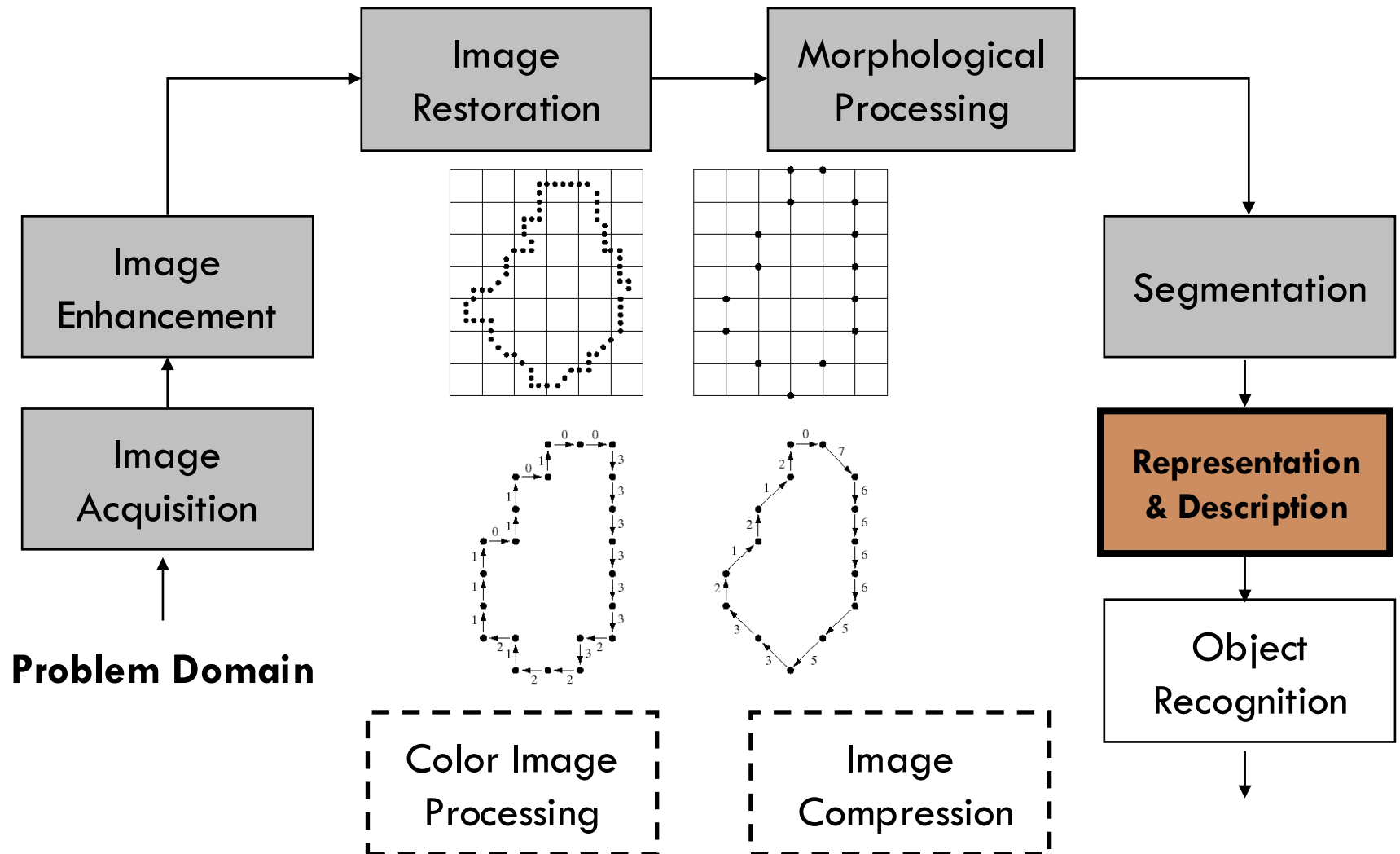
Semantic Segmentation



Instance Segmentation

Key Stages in Digital Image Processing

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Representation and Description

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- ❑ A segmented region can be represented by boundary pixels or by internal pixels
- ❑ Representing region in 2 ways
 - ✓ in terms of its external characteristics (its boundary) $\Rightarrow$ focus on shape characteristics
 - ✓ in terms of its internal characteristics (its region) $\Rightarrow$ focus on regional properties, e.g., color, texture
- ❑ sometimes, we may need to use both ways
- ❑ The description of a region is based on its representation,
 - ✓ for example, a boundary can be described by its length

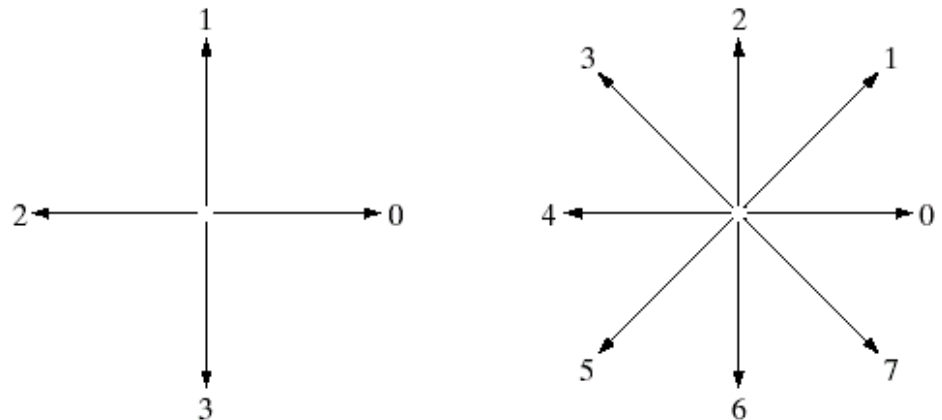
Representation and Description

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- Chain Codes are used to represent the boundary by a connected sequence of straight-line segments of specified length and direction
- Typically, this representation is based on 4 or 8 connectivity of the segments

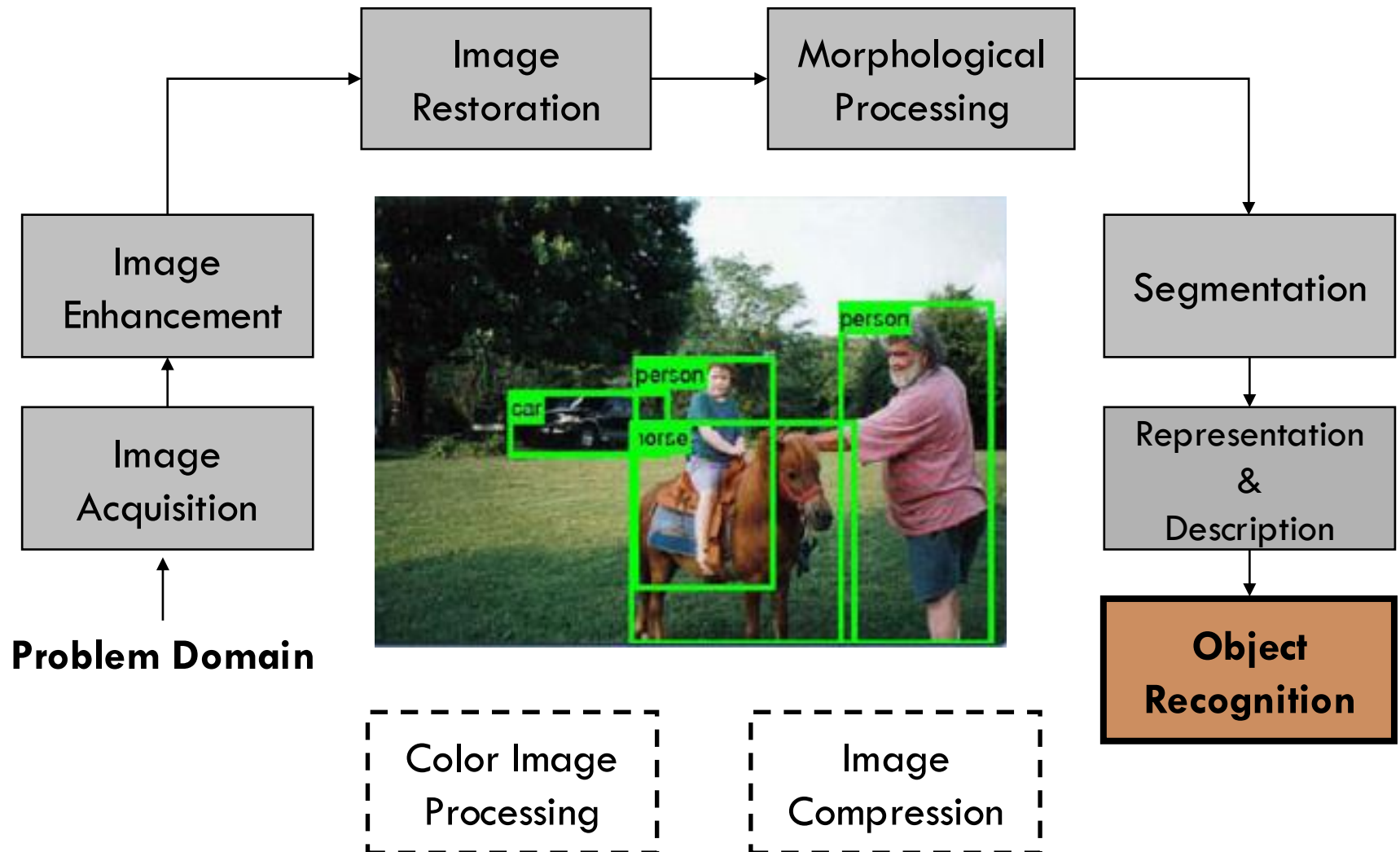
a b

FIGURE 11.1
Direction numbers for
(a) 4-directional
chain code, and
(b) 8-directional
chain code.



Key Stages in Digital Image Processing

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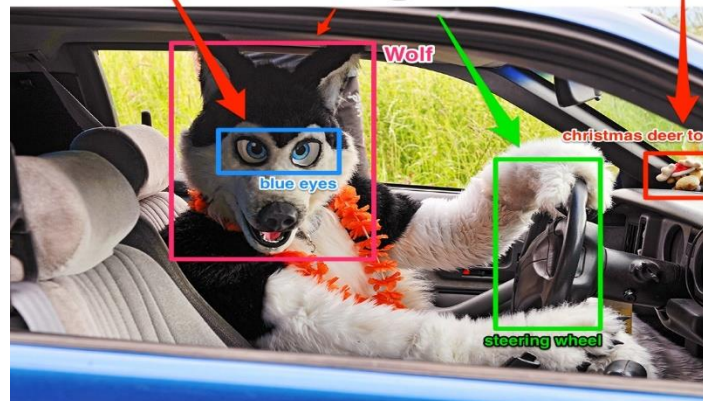


Object Recognition

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- ❑ Recognition is the process that assigns a label to an object based on its descriptors
- ❑ A pattern is an arrangement of descriptors also known as features
- ❑ A pattern class is a family of patterns that share some common properties
- ❑ Pattern recognition by machine involves techniques for assigning patterns to their respective classes automatically and with as little human intervention as possible

Automatic Object Detection in Images & Google



Key Stages in Digital Image Processing

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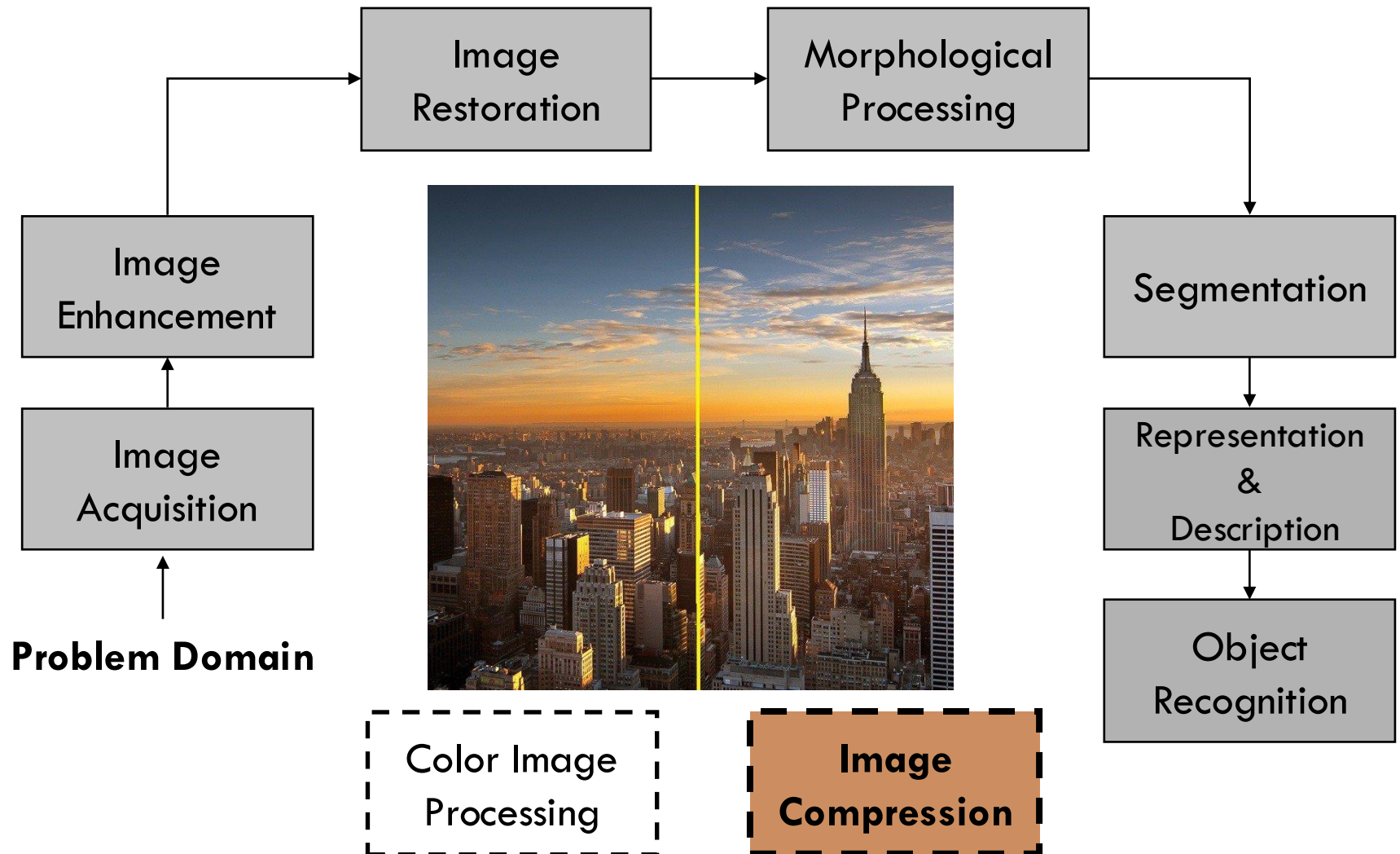


Image Compression

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- ❑ Image compression is minimizing the size in bytes of a graphics file without degrading the quality of the image to an unacceptable level.
- ❑ The reduction in file size allows more images to be stored in a given amount of disk or memory space. It also reduces the time required for images to be sent over the Internet or downloaded from Web pages.
- ❑ Image Compression methods can be based on either:

- ✓ Lossy Compression methods
- ✓ Lossless Compression methods

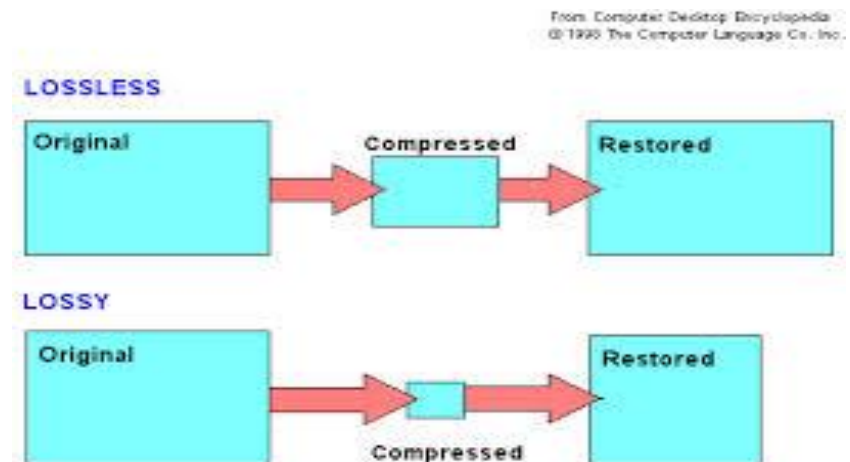


Image Compression

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Original Image (lossless PNG, 60.1 KiB size) — uncompressed is 108.5 KiB



Medium compression (92% less information than uncompressed PNG, 4.82 KiB)



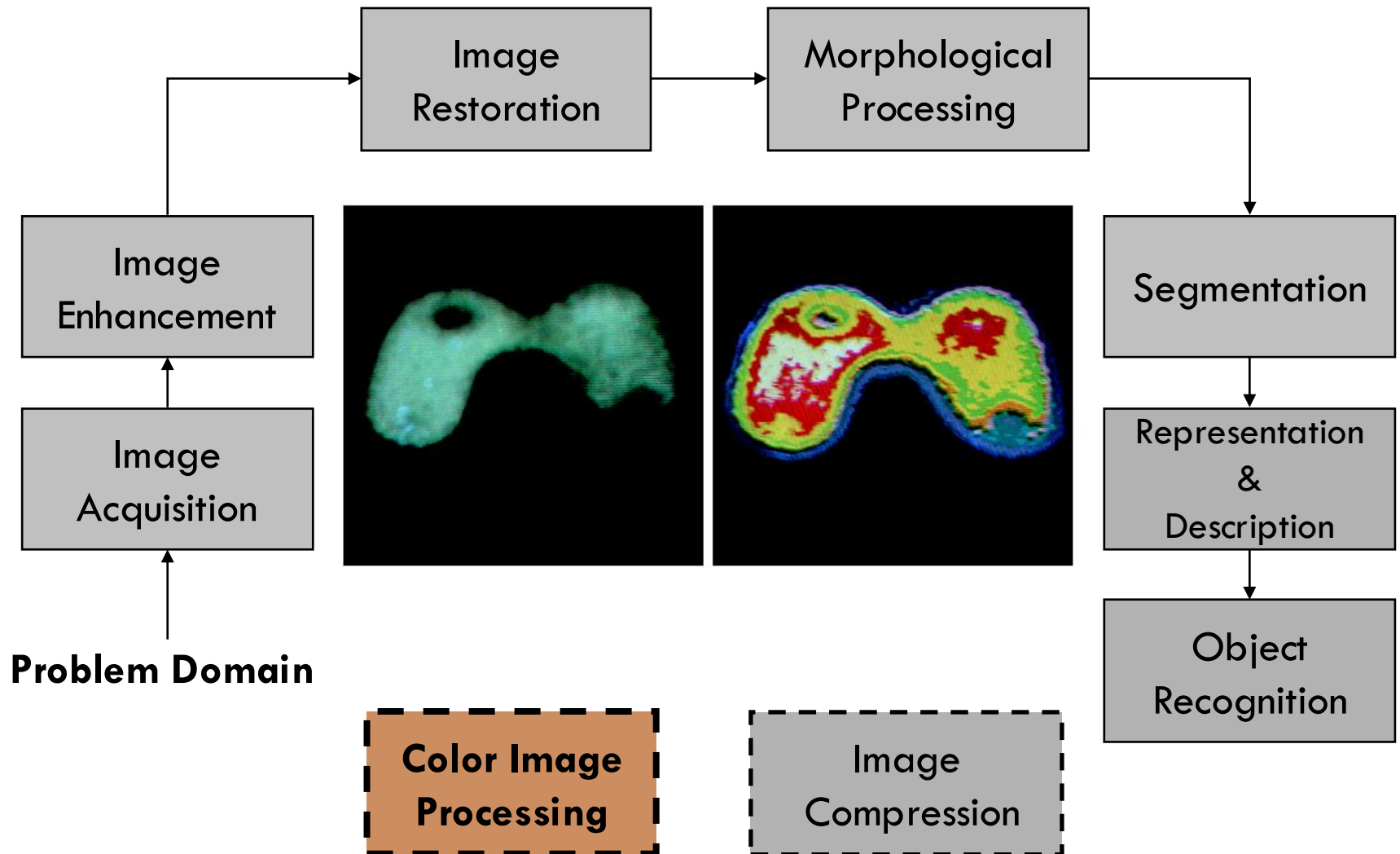
Low compression (84% less information than uncompressed PNG, 9.37 KiB)



High compression (98% less information than uncompressed PNG, 1.14 KiB)

Key Stages in Digital Image Processing

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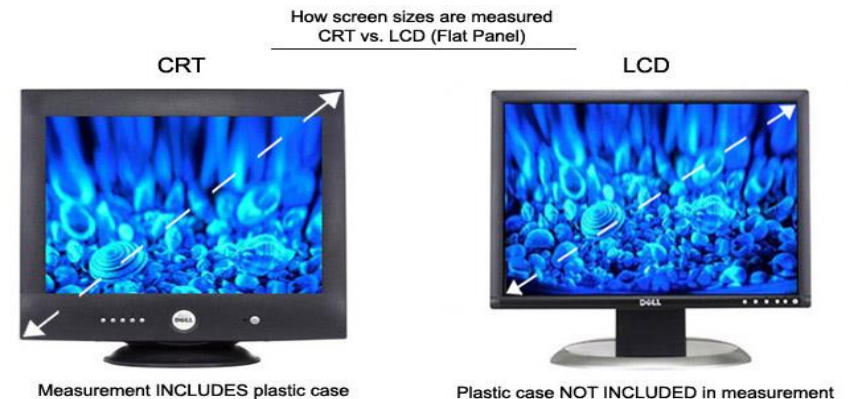
53

Image Processing Components

Image Processing Components

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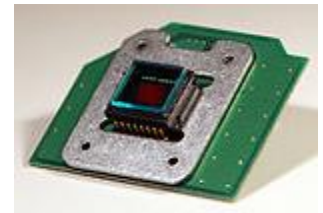
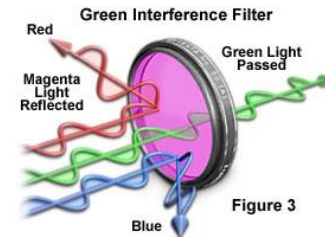
- ❑ Image Sensing device
- ❑ Storage Media
- ❑ Processing Systems
- ❑ Displays [5]
- ❑ Communication Media
- ❑ Hardcopy devices (e.g Printer)
- ❑ Frame Grabber



Camera

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- ❑ Lens (CMount, CSMount) [3-4]
- ❑ Optical Filter (Selectivity in EM waves)
- ❑ Imaging Sensor (CCD Sensor ,CMOS Sensor)[6]
- ❑ Pixel count
- ❑ Optical vs Digital Zoom
- ❑ Flash (Used for lighting/Illumination)



- ✓ Front Illumination
- ✓ Back Illumination

Pixel count [6]

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- ❑ Gross Pixel count
 - ✓ The gross count refers to the total number of pixels on the sensor
- ❑ Effective Pixel count
 - ✓ Effective count tells you how many pixels will be used when taking video or still photos
- ❑ For example, a camera that is 2.1 gross/2.0 effective means that the CCD is comprised of a total of 2.1 million pixels, but the image it captures is actually using only 2.0 million pixels to create the photo.



Optical vs Digital Zoom [7]

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❑ Optical Zoom

- ✓ Optical zoom is when the lens actually moves in and out and gets you closer to the object. An optical zoom is a “real zoom”.

❑ Digital Zoom

- ✓ Digital pictures are made up of tons of tiny dots called pixels. A digital zoom just takes those small pixels and enlarges them internally.



Original



10x Optical



10x Digital

Image Editing Apps [11]

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1. YouCam Perfect
2. YouCam Makeup
3. PhotoDirector
4. Photoshop Express Photo Editor
5. Snapseed
6. Picsart
7. Adobe Lightroom Mobile
8. BeFunky
9. VSCO
10. Instasize Photo Editor
11. Photo Editor Pro - Polish
12. piZap
13. Canva
14. Airbrush
15. Pixlr

The global Image Editing Software market size was valued at USD 1079.52 million in 2022 and is expected to expand at a CAGR of 8.66% during the forecast period, reaching USD 1776.93 million by 2028 [12].

Key Digital Image Processing Companies [13]:

Nvidia Corporation

Canon Inc.

Samsung Electronics Co. Ltd.

Panasonic Corporation

Cognex Corporation

Teledyne Technologies

Vegavid Technology

Visionary.ai.

IBM

Microsoft

Vyro

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4. <http://www.securityideas.com/corcsmount.html>
5. <http://compreviews.about.com/od/multimedia/a/CRTvsLCD.htm>
6. http://camcorders.about.com/od/camcorders101/a/cmos_vs_ccd.htm
7. <http://camcorders.about.com/od/camcorder101/a/optialvsdigital.htm>
8. https://en.wikipedia.org/wiki/Shot_transition_detection
9. [https://en.wikipedia.org/wiki/Dissolve_\(filmmaking\)](https://en.wikipedia.org/wiki/Dissolve_(filmmaking))
10. [https://en.wikipedia.org/wiki/Wipe_\(transition\)](https://en.wikipedia.org/wiki/Wipe_(transition))
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12. <https://www.globenewswire.com/news-release/2023/01/30/2597361/0/en/8-66-Growth-in-Image-Editing-Software-Market-by-2023-2028-Latest-Trends-Key-Players-Types-Applications-Opportunities-Challenges-Risks-Factors-Analysis-Gross-Margin-and-Revenue.html>
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For any query Feel Free to ask



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